## Problem 4

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**Problem 1** *Let* $f(x) = \dfrac{\ln(x)}{x-2}$.

(a) *Evaluate the limit.*

$$\lim_{x\to 2^-} \frac{\ln(x)}{x-2}$$

**Explanation.** This limit is of the form $\dfrac{\#}{0}$, the numerator positive, and the denominator is negative. Therefore, $\displaystyle\lim_{x\to 2^-} \frac{\ln x}{x-2} = -\infty$.

(b) *Find the vertical asymptotes of* $f$. ***EXPLAIN*** *and justify your answer.*

**Explanation.** $x = 2$ is a vertical asymptote of $f$ since $\displaystyle\lim_{x\to 2^-} \frac{\ln x}{x-2} = -\infty$ calculated in part (a).
On the other hand, we know that
$\displaystyle\lim_{x\to 0^+} \ln(x) = -\infty$.
So, it follows that
$\displaystyle\lim_{x\to 0^+} \frac{\ln(x)}{x-2} = \infty$,
since the limit is of the form $\dfrac{\infty}{\#}$, the numerator negative, and the denominator negative. That means $x = 0$ is a vertical asymptote of $f$ since $\displaystyle\lim_{x\to 0^+} \frac{\ln x}{x-2} = \infty$